Scheduling database backups

SnapManager Oracle

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Scheduling database backups

SnapManager (3.2 or later) for Oracle enables you to schedule database backups to occur on a regular basis during off-peak hours to maintain high performance. To schedule a backup, you can create a profile, which includes the database information and retention policy, and then set schedules for the backup.

> You must schedule the backups as an administrator. If you try to schedule the backups as a non-existing user, SnapManager displays an error message: Invalid user: username: Cannot create schedule backup for a given user

The following are some of the schedule-related tasks:

- Schedule a database backup to occur on an hourly, daily, weekly, monthly, or one-time basis.
- View a list of scheduled backups associated with a profile.
- Update a scheduled backup.
- Suspend a schedule temporarily.
- Resume the suspended schedule.
- Delete a schedule.

> The **Run Now Menu Operation** check box is disabled when a scheduled backup is running for that schedule.

Creating backup schedules

You can schedule a backup to occur at the time and frequency that are suited for your data and environment.

From SnapManager 3.2 for Oracle, you can schedule the backups of the archive log files separately. However, you must use the profile that you created to separate the archive log files.

If you have scheduled the backups of the data files and archive log files at the same time, then SnapManager creates the data files backup first.

If you select the schedule interval as -onetimeonly, then all the pruning options are available. If you select a schedule interval other than -onetimeonly, then the pruning options -until-SCN and -until-date are not supported and the following error message is displayed: The archive log pruning option you have specified, -until-scn or -until-date for the schedule interval hourly is invalid. Specify either the -onetimeonly option for the schedule interval, or prune the archive logs using any one of the option all, or -before {-months | -days | -weeks| -hours}.

When a failover happens in Microsoft Windows Server Failover Cluster (WSFC) and Microsoft Cluster Server (MSCS) environments, you must restart the SnapManager for Oracle server so that the service (virtual) address is mapped to the active host and the SnapManager schedules are adjusted to the active SnapManager host.

If the same profile and schedule name exists in another repository, the backup scheduling operation is not initiated in that repository. The operation will exit with the following message: operation is already running.

1. Enter the following command: smo schedule create -profile profile_name {[-full {-online | -offline | -auto}[-retain [-hourly | -daily | -weekly | -monthly | -unlimited] [-verify]] | [-data [-filesfiles [files]] | [-tablespaces-tablespaces [-tablespaces]] {-online | -offline | -auto}[-retain [-hourly | -daily | -weekly | -monthly | -unlimited] [-verify]] | [-archivelogs]} [-commentcomment][-backup-destpath1 [,[path2]] [-exclude-destpath1 [,path2]] [-prunelogs{-all | -untilSCNuntilSCN | -until-dateyyyy-MM-dd HH:mm:ss | -before {-months | -weeks | -days | -hours}} -prune-destprune_dest1,prune_dest2] -schedule-nameschedule_name [-schedule-commentschedule_comment] -interval {-hourly | -daily | -weekly | -monthly | -onetimeonly} -cronstringcronstring-start-time {start-timestart_time <yyyy-MM-dd HH:mm>} -runasuser-runasuser [-force] [-taskspec-taskspec] [-quiet | -verbose]

If you want to…	Then…
Schedule a backup of an online or offline database	Specify -offline or -online to schedule a backup of the offline or online database. If you specify these, you cannot use -auto.
Let SnapManager handle scheduling of a database regardless of whether it is online or offline	Specify -auto. If you specify -auto, you cannot use --offline or -online.
Schedule a backup of data files	Specify -data -files to list the files separated by commas. For example, use file names f1,f2,f3.
Schedule a partial backup of specific tablespaces	Specify -tablespaces to list the tablespaces separated by commas. For example, use ts1,ts2,ts3.
Schedule backup of archive log files	Specify the following: • -archivelogs to schedule backup of the archive log files • -backup-dest to schedule archive log file destinations to be included in the backup • -exclude-dest to schedule the archive log destinations to be excluded from the backup

Specify the retention class values	Specify -retain and indicate whether the backup should be retained according to one of the following retention classes: • -hourly • -daily • -weekly • -monthly • -unlimited SnapManager defaults to hourly.
Schedule pruning of archive log files	Specify the following: **-prunelogs to prune the archive log files while scheduling a backup** -prune-dest to specify the archive log destination from which the archive log files are pruned
Include a name for the schedule	Specify -schedule-name.
Schedule backup of the database at a specific time interval	Specify the interval option and select the time interval from the following, by which the backups should be created: • -hourly • -daily • -weekly • -monthly • -onetimeonly
Configure a schedule	Specify -cronstring and include the following seven subexpressions that describe the individual option: • 1 refers to seconds. • 2 refers to minutes. • 3 refers to hours. • 4 refers to a day in a month. • 5 refers to the month. • 6 refers to a day in a week. • (Optional) 7 refers to the year. **Note:** If you scheduled your backup with different times in -cronstring and -start-time, then the schedule of the backup is overwritten and triggered by the -start-time.
Add a comment about the backup schedule	Specify -schedule-comment followed by the description string.

Specify the start time of the schedule operation	Specify -start-time in the yyyy-mm-dd hh:mm format.
Change the user of the scheduled backup operation while scheduling the backup	Specify -runasuser. The operation runs as the user (root user or Oracle user) who created the schedule. However, you can use your own user ID, if you have valid credentials for both the database profile and host.
Enable a pretask or post-task activity of the backup schedule operation by using the pretask and post-task specification XML file	Specify the -taskspec option and provide the absolute path of the task specification XML file for performing a preprocessing or a post-processing activity to occur before or after the backup schedule operation.

Updating a backup schedule

You can view a list of scheduled operations and update them if necessary. You can update the scheduling frequency, the start time of the schedule, cronstring expression, and the user who scheduled the backup.

1. To update the schedule for a backup, enter this command: smo schedule update -profile profile_name-schedule-nameschedulename [-schedule-commentschedule comment] -interval {-hourly | -daily | -weekly | -monthly | -onetimeonly} -start-timestarttime-cronstringcronstring-runasuserrunasuser [-quiet | -verbose]

Viewing a list of scheduled operations

You can view a list of scheduled operations for a profile.

1. To display information about scheduled operation, enter this command: smo schedule list -profile profile_name[-quiet | -verbose]

Suspending backup schedules

SnapManager enables you to suspend a backup schedule until the backup schedule is resumed.

You can suspend the active schedules. If you try to suspend the backup schedule that is already suspended, you might encounter error message "Cannot suspend: schedule <schedulename> already in suspend state".

1. To suspend the backup schedule temporarily, enter this command: smo schedule suspend -profile profile_name-schedule-nameschedulename [-quiet | -verbose]

Resuming backup schedules

Administrators have the option to resume the suspended backup schedule.

If you try to resume the active schedules, you might encounter the error message: "Cannot resume: schedule

<schedulename> already in resume state".

1. To resume the suspended backup schedule, enter this command: smo schedule resume -profile profile_name-schedule-nameschedulename [-quiet | -verbose]

Deleting backup schedules

You can delete backup schedules when they are no longer necessary.

1. To delete the backup schedule, enter this command: smo schedule delete -profile profile_name-schedule-nameschedulename [-quiet | -verbose]

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